



3889 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Cycle: 2, Proposal Category: GO

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OBSERVATIONS

Folder	Observation	Label	Observing Template	Science Target
EMGW-ToO-BRIGHT-E1				
	1	NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(1) EMGW-TOO-BRIGHT-E1
	2	LRS	MIRI Low Resolution Spectroscopy	(1) EMGW-TOO-BRIGHT-E1
	3	Imaging	MIRI Imaging	(1) EMGW-TOO-BRIGHT-E1
EMGW-ToO-BRIGHT-E2				
	4	NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(2) EMGW-TOO-BRIGHT-E2
	5	LRS	MIRI Low Resolution Spectroscopy	(2) EMGW-TOO-BRIGHT-E2
	6	Imaging	MIRI Imaging	(2) EMGW-TOO-BRIGHT-E2
EMGW-ToO-FAINT				
	7	NIRSPEC	NIRSpec Fixed Slit Spectroscopy	(3) EMGW-TOO-FAINT
	8	Imaging	MIRI Imaging	(2) EMGW-TOO-BRIGHT-E2

ABSTRACT

We now know that binary neutron star mergers are a prolific site of r-process nucleosynthesis where elements heavier than iron are synthesized. However, it is still unknown whether, which and how much of the heaviest of the heavy elements (in the second and third abundance peaks) are synthesized. The key to answering this question is late-time infrared spectroscopy. This is only now possible with the exquisite sensitivity of the

JWST NIRSPEC prism and JWST MIRI. The overlap between JWST Cycle 2 and the upcoming fourth gravitational wave observing run presents an unprecedented opportunity to answer this question. Here, we request three non-disruptive target of opportunity observations of neutron star mergers. Our triggers are non-disruptive as our science goals drive us to wait for the ejecta to become optically thin and wait until the heaviest elements dominate the electron heating. Only then will we be able to directly identify multiple heavy elements in the ejecta and robustly determine relative nucleosynthetic yields.

OBSERVING DESCRIPTION

We request a total of 24.9 hours for 3 non-disruptive Target of Opportunity (ToO) observations with JWST NIRSpec and MIRI to obtain 0.6 - 10 μm spectroscopic and/or photometric follow-up measurements of kilonovae. We request JWST observations to commence between 15-25 days after the ToO trigger is submitted.

Assuming that the Gravitational Wave observing run begins in May 2023 and fully overlaps with Cycle 2, our trigger probability is >80%. Since the gravitational wave duration is 18 months, we request long-term status in Cycle 3.

Given the rapid evolution of kilonovae, we request the special requirement of non-interruptible group observations for the NIRSpec and MIRI observations to get the full wavelength coverage for each ToO.

Given the immense value of this data to the astronomy community, we waive the exclusive proprietary access period so that our data are made publicly available immediately.

Proposal 3889 - Targets - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Generic Targets	#	Name	Criteria	Description
	(1)	EMGW-TOO-BRIGHT-E1	BRIGHT EM-GW Event	
	(2)	EMGW-TOO-BRIGHT-E2	BRIGHT EM-GW Event	
	(3)	EMGW-TOO-FAINT	FAINT EMGW ToO	
	(4)	BRIGHT-E1-TA-OFFSET-STAR	MIRI LRS TA Offset star for BRIGHT-E1 ToO	
	Comments: If we are unable to achieve a high enough SNR on for science target of the EMGW-TOO-BRIGHT-E1 observation for the MIRI LRS target acquisition, we will specify a bright, nearby star to perform a TA offset to the science target.			
	(5)	BRIGHT-E2-TA-OFFSET-STAR	NIRSPEC FS TA Offset star for BRIGHT-E2 ToO	
	Comments: If we are unable to achieve a high enough SNR on for science target of the EMGW-TOO-BRIGHT-E2 observation for the NIRSpec Fixed-Slit target acquisition, we will specify a bright, nearby star to perform a TA offset to the science target.			
	(6)	FAINT-E1-TA-OFFSET-STAR	NIRSPEC FS TA Offset star for FAINT-E1 ToO	
	Comments: If we are unable to achieve a high enough SNR on for science target of the EMGW-TOO-FAINT-E1 observation for the NIRSpec Fixed-Slit target acquisition, we will specify a bright, nearby star to perform a TA offset to the science target.			

Proposal 3889 - Observation 1 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Observation	Proposal 3889, Observation 1: NIRSPEC										Thu May 25 23:02:41 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria									Description
	(1)	EMGW-TOO-BRIGHT-E1	BRIGHT EM-GW Event									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	SUB2048	CLEAR	NRSRAPIDD6	3	1	1	14.452	142230.18	
Template	Slit					Subarray						
	S200A1					FULL						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	3						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	40	1	1	NONE	3	3	1794.433	

Proposal 3889 - Observation 1 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Special Requirements	On Hold on hold for ToO activation Target Of Opportunity Response Time 15 Days, Number of Activations 1, Long-Term ToO Group Observations 1, 2, 3, Non-interruptible
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Proposal 3889 - Observation 2 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Observation	Proposal 3889, Observation 2: LRS Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Generic Targets	#	Name	Criteria	Description					
	(1)	EMGW-TOO-BRIGHT-E1	BRIGHT EM-GW Event						
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SAME	F560W	FASTGRPAVG64	10	1	1	1776.026	142230.26
Template	Subarray			Obtain Verification Image?					
	FULL			true					
Dithers	#	Dither Type		No. Spectral Steps	Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD							
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID
	1	FASTR1	10	1	1	1	1	27.75	F560W

Proposal 3889 - Observation 2 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	400	4	8	1	2	8896.778	
Special Requirements	On Hold on hold for ToO activation								
	Group Observations 1, 2, 3, Non-interruptible								

Proposal 3889 - Observation 3 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Thu May 25 23:02:41 GMT 2023

Observation	Proposal 3889, Observation 3: Imaging										Thu May 25 23:02:41 GMT 2023
	Diagnostic Status: Warning										
	Observing Template: MIRI Imaging										
Diagnostics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria								Description
	(1)	EMGW-TOO-BRIGHT-E1	BRIGHT EM-GW Event								
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets	1	4		1	1	POINT SOURCE	POSITIVE	SMALL	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	80	1	1	Dither 1	4	4	888.013	
	2	F770W	FASTR1	140	1	1	Dither 1	4	4	1554.022	
Special Requirements	On Hold on hold for ToO activation										
	Group Observations 1, 2, 3, Non-interruptible										

Proposal 3889 - Observation 4 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Observation	Proposal 3889, Observation 4: NIRSPEC										Thu May 25 23:02:41 GMT 2023	
	Diagnostic Status: Warning											
	Observing Template: NIRSpec Fixed Slit Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria									Description
	(2)	EMGW-TOO-BRIGHT-E2	BRIGHT EM-GW Event									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	CLEAR	NRSRAPIDD6	3	1	1	171.788	142230.23	
Template	Slit					Subarray						
	S200A1					FULL						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	100	1	1	NONE	5	5	7367.389	

Proposal 3889 - Observation 4 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Special Requirements	On Hold on hold for ToO activation Target Of Opportunity Response Time 15 Days, Number of Activations 1, Long-Term ToO Group Observations 4, 5, 6, Non-interruptible
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Proposal 3889 - Observation 5 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Observation	Proposal 3889, Observation 5: LRS Diagnostic Status: Warning Observing Template: MIRI Low Resolution Spectroscopy								
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.								
Generic Targets	#	Name	Criteria	Description					
	(2)	EMGW-TOO-BRIGHT-E2	BRIGHT EM-GW Event						
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	SAME	F560W	FASTGRPAVG64	10	1	1	1776.026	142230.25
Template	Subarray				Obtain Verification Image?				
	FULL				true				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps	
	1	ALONG SLIT NOD							
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID
	1	FASTR1	10	1	1	1	1	27.75	F560W

Proposal 3889 - Observation 5 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	400	4	8	1	2	8896.778	
Special Requirements	On Hold on hold for ToO activation								
	Group Observations 4, 5, 6, Non-interruptible								

Proposal 3889 - Observation 6 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Observation	Proposal 3889, Observation 6: Imaging										Thu May 25 23:02:41 GMT 2023
	Diagnostic Status: Warning Observing Template: MIRI Imaging										
Diagnostics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria				Description				
	(2)	EMGW-TOO-BRIGHT-E2	BRIGHT EM-GW Event								
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets	1	4		1	1	POINT SOURCE	POSITIVE	SMALL	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	140	1	1	Dither 1	4	4	1554.022	
	2	F770W	FASTR1	140	1	1	Dither 1	4	4	1554.022	
Special Requirements	On Hold on hold for ToO activation										
	Group Observations 4, 5, 6, Non-interruptible										

Proposal 3889 - Observation 7 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Observation	Proposal 3889, Observation 7: NIRSPEC										Thu May 25 23:02:41 GMT 2023	
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec Fixed Slit Spectroscopy											
	(Visit 7:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Generic Targets	#	Name	Criteria									Description
	(3)	EMGW-TOO-FAINT	FAINT EMGW ToO									
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	SAME	WATA	FULL	CLEAR	NRSRAPIDD6	3	1	1	171.788	142230.24	
Template	Slit					Subarray						
	S200A1					FULL						
Dithers	#	Primary Dither Positions						Sub-Pixel Pattern				
	1	5						NONE				
Spectral Elements	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSIRS2RAPID	100	1	1	NONE	5	5	7367.389	

Proposal 3889 - Observation 7 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Special Requirements	On Hold on hold for ToO activation Target Of Opportunity Response Time 15 Days, Number of Activations 1, Long-Term ToO Group Observations 7, 8, Non-interruptible
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Proposal 3889 - Observation 8 - Identifying the heaviest of the heavy elements synthesized in neutron star mergers

Observation	Proposal 3889, Observation 8: Imaging										Thu May 25 23:02:41 GMT 2023
	Diagnostic Status: Warning Observing Template: MIRI Imaging										
Diagnostics	(Visit 8:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Generic Targets	#	Name	Criteria				Description				
	(2)	EMGW-TOO-BRIGHT-E2	BRIGHT EM-GW Event								
Template	Subarray										
	FULL										
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size	
	1	4-Point-Sets	1	4		1	1	POINT SOURCE	POSITIVE	SMALL	
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	F560W	FASTR1	300	1	1	Dither 1	4	4	3330.048	
Special Requirements	On Hold on hold for ToO activation										
	Group Observations 7, 8, Non-interruptible										